

Joint Client Selection and Bandwidth Allocation Algorithm for Federated Learning

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Abstract—In federated learning (FL), if the participating mobile devices have low computing power and poor wireless channel conditions and/or they do not have sufficient data for various classes, a long convergence time is required to achieve the desired model accuracy. To address this problem, we first formulate a constrained Markov decision process (CMDP) problem that aims to minimize the average time of rounds while maintaining the numbers of trained data and trained data classes above certain numbers. To obtain the optimal scheduling policy, the formulated CMDP problem is converted into an equivalent linear programming (LP). Additionally, to overcome the problem of the curse of dimensionality in CMDP, we develop a joint client selection and bandwidth allocation algorithm (J-CSBA) that jointly selects appropriate mobile devices and allocates suitable amount of bandwidth to them at each round by considering their data information, computing power, and channel gain. Evaluation results validate that J-CSBA can reduce the convergence time by up to 49% compared to a conventional random scheme.

Index Terms—Federated learning (FL), joint optimization, client selection, bandwidth allocation, convergence time, constrained Markov decision process (CMDP)

1 INTRODUCTION

To alleviate the privacy issue of data owners and facilitate collaborative deep learning among distributed clients, federated learning (FL), which is one of decentralized machine learning approaches, has been introduced [1]. In FL for image classification, clients (e.g., mobile devices) are randomly selected as participants by an FL server. Then, the FL server multicasts a global model to these clients. Then, the clients update the model based on their local data. After finishing the model update, they upload their updated parameters to the FL server. Finally, the FL server aggregates the parameters from multiple clients and repeats these steps (a.k.a., round), until a desired model accuracy is achieved. However, when mobile devices are used as clients, random selection of participants leads to the performance degradation.

For example, if some mobile devices (i.e., stragglers) with low computing power and poor wireless channel conditions are selected, they cannot return the updated parameters in a timely manner. That is, the FL server cannot complete a round within a short duration, which leads a long convergence time. In such a situation, more bandwidth should be allocated to the selected mobile devices with low computing power and/or poor wireless channel conditions.

Meanwhile, when the data classes of the selected mobile devices are unevenly distributed (e.g., the selected mobile devices have only partial classes), the desired accuracy of the model cannot be achieved even after a certain number of rounds [2]. In other words, to achieve a desired accuracy of the model, the selected mobile devices should have near-independent and identically distributed (near-IID) data. That is, the selected mobile devices should have as various classes as possible. At the same time, a sufficient number of data should be collected from mobile devices to provision short convergence time of FL. To summarize, to achieve high accuracy within a short duration, mobile devices with sufficient number of data in diverse classes should be selected in each round. At the same time, more bandwidth should be allocated to mobile devices with low computing power and/or poor wireless channel conditions. However, there has been no previous work on jointly optimizing client selection and bandwidth allocation.

In this paper, to minimize the average time of rounds while maintaining the numbers of trained data per round and trained data classes per round above certain numbers, a constrained Markov decision process (CMDP) problem is formulated. To obtain the optimal scheduling policy, the formulated CMDP problem is converted into an equivalent linear programming (LP). To overcome the problem of the curse of dimensionality in CMDP, we propose a joint client selection and bandwidth allocation algorithm (J-CSBA)

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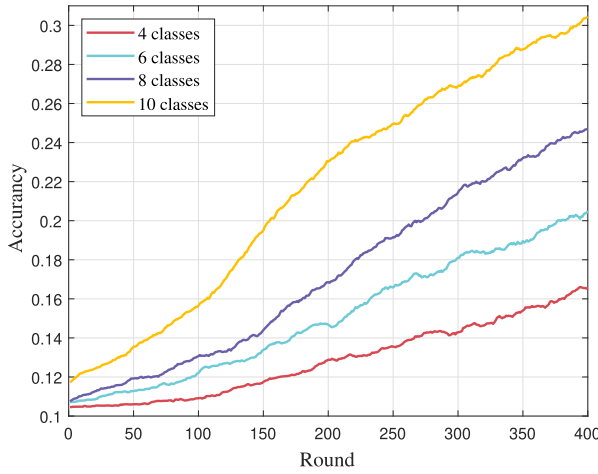


Fig. 1. Achievable accuracy as a function of round according to the number of trained classes per round.

which can be implemented as one of applications in the open radio access network (O-RAN) architecture [3]. Through the interfaces of O-RAN, the information (e.g., the numbers of data and data classes) of mobile devices is collected for the operation of J-CSBA. After that, for each round, J-CSBA jointly selects appropriate mobile devices and allocates a relevant amount of bandwidth to them at each round by considering their data information, computing power, and channel gain. Evaluation results demonstrate that J-CSBA can reduce the average latency to achieve a desired accuracy by up to 49% compared to a random scheme. In addition, it can be shown that J-CSBA adaptively selects mobile devices and allocates appropriate amounts of bandwidth to them according to the system environments.

The contributions of this paper can be summarized as follows: 1) the joint optimization of client selection and bandwidth allocation can be performed by J-CSBA, which minimizes the convergence time to achieve a certain accuracy of the model; 2) J-CSBA is implementation-friendly since it has linear complexity and can operate over the promising O-RAN architecture; and 3) the presented evaluation results under various environments are analyzed, which can be exploited for designing an FL system in mobile networks.

The remainder of this paper is organized as follows. Motivating examples are described in Section 2. Related works are summarized in Section 3. The system model is presented in Section 4, and the CMDP model is formulated in Section 5. Then, J-CSBA is developed in Section 6, and the evaluation results are discussed in Section 7. Finally, the concluding remarks are given in Section 8.

2 MOTIVATING EXAMPLE

In this section, based on the preliminary simulation results, we will show that the selected mobile devices should have as various classes as possible and a sufficient number of data to provision short convergence time of FL.

Fig. 1 shows the achievable accuracy after certain rounds as the number of trained classes per round increases. In this result, we use the CIFAR-10 dataset [4], which is an object classification dataset consisting of 50,000 training images and 10,000 testing images in 10 classes. In addition, each mobile device has 100 images in 2 classes, and the FL server

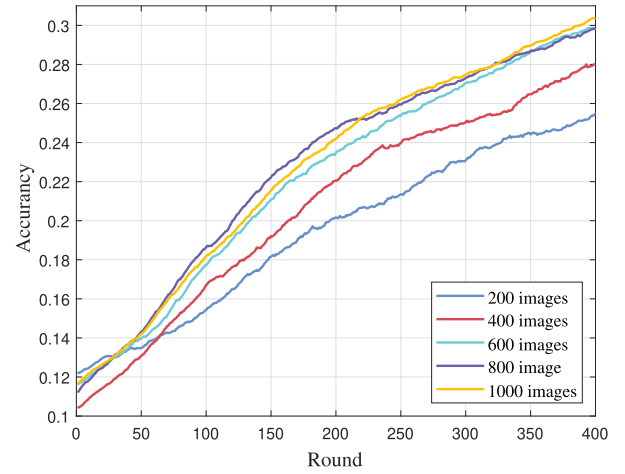


Fig. 2. Achievable accuracy as a function of round according to the number of trained data per round.

selects a fixed number of mobile devices (i.e., 6)¹ to make the number of trained classes per round a certain number (i.e., 4, 6, 8, and 10). From Fig. 1, it can be shown that the achievable accuracy tends to increase with the number of trained classes per round. This indicates that the selected mobile devices should have as various classes as possible to achieve a high accuracy within a short duration.

Fig. 2 shows the achievable accuracy after certain rounds for different numbers of trained data per round. In this result, each mobile device has 100 images in 5 classes, and the FL server selects mobile devices such that the number of trained data per round reaches a certain number (i.e., 200, 400, 600, 800, and 1000). Note that, for all cases, the number of trained classes per round is fixed at 10. From Fig. 2, it can be seen that a sufficient number of trained data per round (e.g., over 600) is required to achieve high accuracy after certain rounds.

To summarize, the numbers of trained data per round and the trained data classes per round should be maintained above certain numbers to achieve the reduced convergence time in FL.

3 RELATED WORK

To reduce the convergence time in FL, several studies have been conducted in the literature. These studies can be classified into: 1) data distribution adjustment [5], [6], [7], [8]; 2) model compression [9], [10], [11], [12], [13]; and 3) client selection/scheduling [14], [15], [16], [17], [18], [19], [20], [21], [22].

Zhao *et al.* [5] developed a data sharing algorithm where the FL server has the shared data and sends some data to the clients to make the data distribution of the clients near-IID. Chen *et al.* [7] designed a semi-federated learning algorithm where mobile devices are grouped into several clusters, and the mobile devices in each group update the parameters in a sequential manner to mitigate non-IID data distribution. Duan *et al.* [6] introduced a re-balancing algorithm that augments data of a minority class in mobile devices. In addition, they proposed a re-scheduling algorithm that selects mobile devices to make their aggregated data distribution near-IID and commands them to train the

1. The fixed number of mobile devices mean the same number of trained data in each case.

model in a sequential manner. Chiu *et al.* [8] proposed a model exchanging algorithm that swaps the model among the mobile devices in each round, which leads the same effect of exchanging data.

Konečný *et al.* [9] developed two communication-efficient algorithms to reduce uplink latency. One is to upload the full model gradient after parameter quantization, while the other is to upload the randomly-selected or low-rank part of the model. Similarly, Caldas *et al.* [10] proposed a federated dropout method in which each mobile device trains a smaller sub-model instead of the entire global model. Jiang *et al.* [11] proposed a two-stage distributed pruning scheme that employs an initial pruning of trusted clients having high computational power and further pruning during the FL procedure. Chen *et al.* [12] suggested a layer-aware asynchronous model update algorithm that divides the model into shallow layers and deep layers. Then, the algorithm updates the gradients of the deep layers more frequently than those of the shallow layers based on the fact that the gradients of the deep layers change less than those of the shallow layers during training. Mills *et al.* [13] designed a model compression algorithm in which mobile devices send the selected parameters and their encoded indexes after training.

Yang *et al.* [14] proposed an age-based client scheduling algorithm where mobile devices are scheduled by considering the elapsed time from the previous update and their channel quality. Wang *et al.* [15] suggested a deep reinforcement learning-based client selection algorithm, in which the FL server learns how to select a set of participating mobile devices by considering the data distribution. Yoshida *et al.* [16] introduced a multi-armed bandit-based client selection algorithm that reflects the computation and communication capability of clients to avoid the poor-resource clients' participation. Wang *et al.* [17] developed a reputation-based client selection algorithm considering the model score of each client to improve the model accuracy. Huang *et al.* [18] formulated a client selection problem to guarantee fair participation, which can lead to higher accuracy. Yu *et al.* [19] formulated a joint client selection and resource management scheme to maximize the number of selected clients while minimizing their total energy consumption. Zhang *et al.* [20] a weight divergence-based client selection algorithm to frequently select the clients having near-IID data. Tang *et al.* [21] suggested a correlation-based client selection algorithm that selects clients having high data correlation with other clients to effectively decrease the posterior expectation of the overall loss in the model. Song *et al.* [22] formulated a joint optimization problem on the uplink/downlink transmission duration and computation rates of clients to minimize the round time.

However, no work has been conducted on the joint optimization of client selection and bandwidth allocation. Therefore, sufficiently short convergence time cannot be guaranteed, especially when a given bandwidth is not sufficient.

4 SYSTEM MODEL

Fig. 3 shows the system model in this paper. There are N mobile devices in the coverage of base station (BS). It is assumed that the FL server is located at the edge cloud

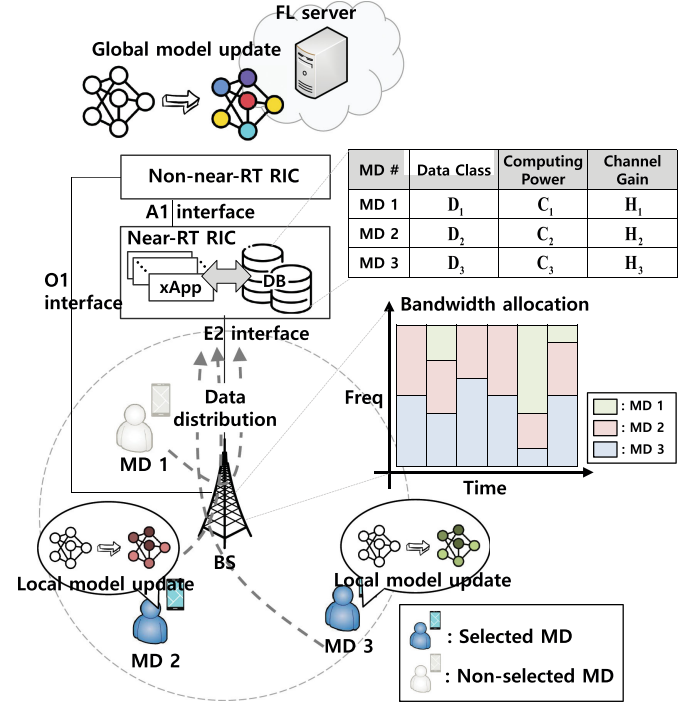


Fig. 3. System model.

attached to BS. In addition, the FL server aims to make a model whose parameter size is W MByte [23] and the required computation power is R floating point operations (FLOPs) in a distributed manner.

Meanwhile, we assume the open radio access network (O-RAN) architecture [3] that integrates different kinds of modular base station applications on the off-the-shelf hardware. The O-RAN architecture includes 1) non-real-time RAN intelligent controller (non-RT RIC) that conducts non-real-time decision (e.g., policy management and analytics) through the O1 interface and 2) near-real-time RAN intelligent controller (near-RT RIC) that conducts near-real-time control and optimization (e.g., resource allocation and load balancing) through the E2 interface.

For optimal client selection and bandwidth allocation, the near-RT RIC collects some information of mobile devices such as the amount of data F , data class D , channel gain H , and available computing power C (in units of floating point operations per second (FLOPs) [24]) through the E2 interface, and the information is stored in the RIC database. Note that, because these information can be expressed with a small number of bits, the signalling overhead to collect the information is not quite high. For example, the number of data, channel gain, and available computing power can be expressed with integer or floating point data types of 4 bytes, and data classes can be expressed by several bits (e.g., 10 bits for CIFAR-10 dataset).² Specifically, if a certain mobile device has the 1st, 3rd, and 6th classes for CIFAR-10 dataset, its data classes can be represented as $D = [1, 0, 1, 0, 0, 1, 0, 0, 0]$. Based on this information, the application (i.e., J-CSBA) installed in the near-RT RIC selects mobile

2. Even considering the IP header overhead, the overall packet size for the information is around 40 bytes. If there are 30 mobile devices participating into the FL procedure, the total signalling overhead is only 1200 bytes/round.

devices and then allocates bandwidths to them.³ Specifically, J-CSBA selects mobile devices with a sufficient amount of data and diverse data classes, because a small amount of trained data and the absence of data classes per round can degrade the performance of the model (e.g., accuracy) (see Figs. 1 and 2). Next, to complete a round within a short duration, J-CSBA allocates an appropriate portion of the bandwidth to each mobile device⁴ by considering the available computing power and channel gain.

The round time T is determined by the slowest mobile device in set $\mathbf{A}_S$ of selected mobile devices in that round. That is, T can be calculated as $T = \max_{i \in \mathbf{A}_S} \{T_B + T_{T,i} + T_{U,i}\}$, where T_B is the latency of multicasting parameters from the FL server to the selected mobile devices. In addition, $T_{T,i}$ and $T_{R,i}$ denote the latency that mobile device i trains the model based on its local data and the latency that mobile device i returns the update parameters to the FL server, respectively. These latencies (i.e., T_B , $T_{T,i}$, and $T_{R,i}$) can be estimated as follows. First, because the transmission power density of BS is much higher than that of mobile devices and it is assumed that BS exploits the entire bandwidth to broadcast the model, T_B can be neglected (i.e., $T_B \approx 0$). In addition, $T_{T,i}$ can be calculated as R/C_i ,⁵ where C_i denotes the computing power of mobile device i . Meanwhile, when B and A_i denote the total bandwidth and the bandwidth allocation ratio to mobile device i , respectively, the allocated bandwidth to mobile device i can be obtained by $A_i B$. Then, the transmission rate of uploading parameters can be calculated as $A_i B \log(1 + \frac{P_M H_i^2}{N_O})$, where N_O and P_M denote the noise power and transmission power density of mobile device, respectively. In addition, H_i is the channel gain of mobile device i . Thus, $T_{R,i}$ can be calculated by $\frac{W}{A_i B \log(1 + \frac{P_M H_i^2}{N_O})}$.

Because the round time T is determined by the slowest mobile device selected in that round, to minimize round time T , the application (i.e., J-CSBA) installed in the near-RT RIC allocates a larger portion of the bandwidth to the mobile devices with lower computing powers and worse channel conditions. Note that, for jointly optimizing mobile device selection and bandwidth allocation, we formulate the CMDP model, which will be described in the next section. Moreover, these two kinds of optimization can be handled by one dimension of action space, i.e., the bandwidth allocation ratio, because an action representing zero allocation ratio for a specific mobile device can be interpreted as a situation where that mobile device is not selected.

3. Different types of external applications (xApp) can operate in the O-RAN architecture [3]. These applications can define a set of messages that near-RT RIC and BS exchange through the E2 interface [25]. In addition, the E2 interface can handle virtualized functionalities of BS (i.e., central unit (CU), distributed unit (DU), and radio unit (RU)) [25]. Therefore, in J-CSBA, the client selection module and the bandwidth modules can be implemented at CU (for the radio resource control layer with a long timescale) and DU (for the MAC layer with a short timescale), respectively.

4. We assume orthogonal frequency-division multiplexing (OFDM).

5. Note that the required computation power of the model, R , depends on the number of parameters in the deep network and number of required local trainings (e.g., epoch).

5 CMDP

To select appropriate mobile devices and allocate bandwidth to them, we formulate a CMDP model, where the agent conducts successive appropriate actions to minimize (or maximize) the cost (or reward) under the defined constraints [26]. That is, in this paper, the application installed in the near-RT RIC selects appropriate mobile devices and allocates bandwidth to them at time epoch $T = \{1, 2, 3, \dots\}$ to minimize the round time while maintaining the number of the trained data per round and the number of their classes per round above certain numbers.

5.1 State Space

The state space $\mathbf{S}$ can be defined as

$$\mathbf{S} = \prod_i \mathbf{C}_i \times \mathbf{H}_i \times \mathbf{D}_i \times \mathbf{F}_i \quad (1)$$

where $\mathbf{C}_i$ and $\mathbf{H}_i$ denote the computing power state and channel gain state for mobile device i , respectively. In addition, $\mathbf{D}_i$ represents the data class state for mobile device i . $\mathbf{F}_i$ describes the number of data in mobile device i . Additionally, $\times$ is a Kronecker product.

$\mathbf{C}_i$ can be described by

$$\mathbf{C}_i = \{C_i^1, C_i^2, \dots, C_i^{\max}\} \quad (2)$$

where $C_i^{\max}$ denotes the maximum computing power of mobile device i .

Meanwhile, when the channel gain model is discretized, $\mathbf{H}_i$ can be defined as

$$\mathbf{H}_i = \{H_i^1, H_i^2, \dots, H_i^{\max}\} \quad (3)$$

Note that H_i^1 and $H_i^{\max}$ denote the lowest and highest channel gains, respectively.

Because it is assumed that each mobile device has different subsets of random classes, $\mathbf{D}_i$ can be represented as

$$\mathbf{D}_i = \{D_i^1, D_i^2, \dots, D_i^{N_P}\} \quad (4)$$

where N_P is the number of possible combinations of subsets. In addition, D_i^j is the j th element of $\mathbf{D}_i$. Because we consider the model having K classes (e.g., 10 classes in CIFAR-10 dataset), D_i^j can be denoted as

$$D_i^j = \{d_i^1, d_i^2, \dots, d_i^K\} \quad (5)$$

where d_i^l denotes whether mobile device i has the l th class or not. That is, if $d_i^l = 1$, mobile device i has the l th class. Otherwise, it does not have the l th class.

$\mathbf{F}_i$ can be described by

$$\mathbf{F}_i = \{1, 2, \dots, F_i^{\max}\} \quad (6)$$

where $F_i^{\max}$ denotes the maximum number of data that mobile device i has.

5.2 Action Space

The action space $\mathbf{A}$ can be defined by

$$\mathbf{A} = \prod_i \mathbf{A}_i \quad (7)$$

where $\mathbf{A}_i$ is the action space for mobile device i , which can be denoted by

$$\mathbf{A}_i = \{0, A_i^1, A_i^2, \dots, 1\} \quad (8)$$

where A_i denotes the bandwidth allocation ratio to mobile device i .⁶ In addition, A_i can represent whether mobile device i is selected or not. Specifically, $A_i = 0$ denotes the situation where mobile device i is not selected. Note that the summation of the bandwidth allocation ratios (i.e., actions) to selected mobile devices should be 1 (i.e., $\sum_i \mathbf{A}_i = 1$).

5.3 Transition Probability

Since the state transitions of each mobile device are independent of each other, the transition probability with the chosen action A from the current state S to the next state S' can be represented as

$$P[S'|S, A] = \prod_i P[S'_i|S_i] \quad (9)$$

where S_i and S'_i denote the current and next states of mobile device i , respectively.

All states of mobile device i are not influenced by the chosen action A . In addition, all the states change independently of each other. Thus, the transition probability of the states of mobile device i can be represented as (10), at the top of the next page,

$$P[S'_i|S_i, A] = P[C'_i|C_i] \times P[H'_i|H_i] \times P[D'_i|D_i] \times P[F'_i|F_i] \quad (10)$$

where C_i , H_i , D_i , and F_i are the current computing power state, channel status state, data class state, and data number state, respectively. In addition, C'_i , H'_i , D'_i , and F'_i denote the next computing power state, channel status state, data class state, and data number state, respectively.

$P[C'_i|C_i]$ and $P[H'_i|H_i]$ can be defined in a statistical manner. Meanwhile, in this paper, it is assumed that the data of mobile devices do not change until the current model converges. That is, $P[D'_i|D_i]$ and $P[F'_i|F_i]$ can be defined as

$$P[D'_i|D_i] = \begin{cases} 1, & \text{if } D'_i = D_i \\ 0, & \text{otherwise} \end{cases} \quad (11)$$

and

$$P[F'_i|F_i] = \begin{cases} 1, & \text{if } F'_i = F_i \\ 0, & \text{otherwise.} \end{cases} \quad (12)$$

6. Because the system can allocate bandwidth based on a basic unit (e.g., resource block in LTE system) [27], the action space can be constructed in a discrete manner.

5.4 Cost & Constraint Functions

5.4.1 Cost Function

The round time T is determined by the slowest mobile device selected in that round (i.e., $T = \max_{i \in \mathbf{A}_S} \{T_{B,i} + T_{T,i} + T_{U,i}\}$). Because the objective is to minimize the average time of rounds, the cost function can be defined as

$$r(S, A) = \max_{i \in \mathbf{A}_S} \{T_B + T_{T,i} + T_{U,i}\}. \quad (13)$$

5.4.2 Constraint Functions

To maintain the number of trained data in a single round above a certain number, we define the constraint function on the number of trained data as

$$c_M(S, A) = \sum_i \delta[A_i \neq 0] F_i. \quad (14)$$

Meanwhile, to maintain the number of classes that the selected mobile devices have in a single round above a certain number, the ratio between the number of data classes learned in the current round and the number of whole data classes is exploited as the constraint function for the number of classes, $c_C(S, A)$. That is, $c_C(S, A)$ can be defined as

$$c_C(S, A) = \frac{\sum_t \delta\left[\left(\sum_{i \in \mathbf{A}_S} d_i^t\right) \neq 0\right]}{K}. \quad (15)$$

5.5 Optimization Formulation

The average time of rounds, ζ_L , can be defined as

$$\zeta_L = \limsup_{t \rightarrow \infty} \frac{1}{t} \sum_{t'}^t E[r(S_{t'}, A_{t'})] \quad (16)$$

Meanwhile, the average number of trained data in a round ψ_M and average ratio ψ_C between the number of data classes learned in the round and the number of whole data classes can be represented by

$$\psi_M = \limsup_{t \rightarrow \infty} \frac{1}{t} \sum_{t'}^t E[c_M(S_{t'}, A_{t'})] \quad (17)$$

and

$$\psi_C = \limsup_{t \rightarrow \infty} \frac{1}{t} \sum_{t'}^t E[c_C(S_{t'}, A_{t'})]. \quad (18)$$

The CMDP model can be expressed as follows:

$$\min_{\pi} \zeta_L \quad (19)$$

$$\text{s.t. } \psi_M \geq \theta_M \text{ and } \psi_C \geq \theta_C \quad (20)$$

where π is the policy that represents the probabilities of taking a specific action at a certain state. Moreover, θ_M and θ_C denote the lower bounds on the number of trained data in the round and on the ratio between the number of data classes learned in the round and the number of whole data classes, respectively.

If $\phi(S, A)$ denotes the stationary probability of state S and action A , the equivalent LP model with the formulated CMDP model can be described as

$$\min_{\phi(S,A)} \sum_S \sum_A \phi(S, A) r(S, A) \quad (21)$$

under the following constraints:

$$\sum_S \sum_A \phi(S, A) c_M(S, A) \geq \theta_M \quad (22)$$

$$\sum_S \sum_A \phi(S, A) c_C(S, A) \geq \theta_C \quad (23)$$

$$\sum_A \phi(S', A) = \sum_S \sum_A \phi(S, A) P[S'|S, A] \quad (24)$$

$$\sum_S \sum_A \phi(S, A) = 1 \quad (25)$$

and

$$\phi(S, A) \geq 0. \quad (26)$$

The objective function in (21) aims to minimize the average time of rounds. The number of trained data in a round can be maintained above θ_M with the constraint in (22). The ratio between the number of data classes learned in the round and the number of whole data classes can be maintained above θ_C with the constraint in (23). The Chapman-Kolmogorov equation can be satisfied with the constraint in (24). The probability properties are maintained by the constraints in (25) and (26).

When solving the LP problem, we can obtain the optimal stochastic policy $\pi^*(S, A)$ as the solution of the CMDP model. The optimal stochastic policy means that an action A at state S is selected based on the optimal probability distribution. Because the optimal stationary probability $\phi^*(S, A)$ can achieve the objective of the formulated CMDP model under the constraints, the optimal stochastic policy $\pi^*(S, A)$ can be obtained as

$$\pi^*(S, A) = \frac{\phi^*(S, A)}{\sum_{A'} \phi^*(S, A')} \text{ for } S \in \mathbf{S}, \sum_{A'} \phi^*(S, A') > 0. \quad (27)$$

When $\sum_{A'} \phi^*(S, A') = 0$, which implies that there is no solution that can satisfy all constraints, all mobile devices are selected.

6 J-CSBA

To avoid the curse of dimensionality problem in CMDP, we propose J-CSBA as shown in *Algorithm 1*. In the first loop, the algorithm selects the mobile devices based on the expected performance (i.e., computing power and channel gain) and the possibility that they can contribute to maintain the number of trained classes in each round above a certain number. In addition, in the second loop, the algorithm selects more mobile devices with the highest performance to maintain the number of trained data in each round above a

certain number while keeping that the round time does not become too long. Meanwhile, in the last loop, the algorithm allocates some amount of bandwidth to minimize the round time. The detailed procedure of *Algorithm 1* is as follows.

Algorithm 1. J-CSBA

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1: Initialize the sets  $\mathbf{A}_S$  and  $\mathbf{A}_P$ 
2:  $k \leftarrow 0$ 
3: repeat
4:    $k \leftarrow k + 1$ 
5:    $i^* \leftarrow \arg \min_{i \in \mathbf{A}_P} T_{L,i}$  with the assumption that each mobile
     device exploits the whole bandwidth  $B$ 
6:   if  $(\bigcup_{i \in \mathbf{A}_S} D_i) - D_{i^*} \neq \emptyset$  then
7:     Add mobile device  $i^*$  to  $\mathbf{A}_S$ 
8:     Assign index  $k$  to mobile device  $i^*$ 
9:   end if
10:  Remove mobile device  $i^*$  from  $\mathbf{A}_P$ 
11: until  $\frac{\prod_{i \in \mathbf{A}_S} |D_i|}{K} \geq \theta_C$  or  $\mathbf{A}_P = \emptyset$ 
12: repeat
13:    $i^* \leftarrow \arg \min_{i \in \mathbf{A}_P} T_{L,i}$  with the assumption that each mobile
     device exploits the whole bandwidth  $B$ 
14:   Add mobile device  $i^*$  to  $\mathbf{A}_S$ 
15:   Remove mobile device  $i^*$  from  $\mathbf{A}_P$ 
16: until  $|\mathbf{A}_S| = \theta_M$  or  $\mathbf{A}_P = \emptyset$ 
17:  $k \leftarrow 0$ 
18: repeat
19:    $k \leftarrow k + 1$ 
20:   Allocate  $\frac{T_{L,k}}{\sum_{k'} T_{L,k'}}$   $B$  to mobile device with index  $k$ .
21: until  $k = |\mathbf{A}_S|$ 
    
```

First, $\mathbf{A}_S$ and $\mathbf{A}_P$ are created as an empty set and a set of all the mobile devices, respectively (line 1 in *Algorithm 1*). Then, the mobile device i^* with the shortest individual round time $T_{L,i}$ among the mobile devices in $\mathbf{A}_P$ is chosen (line 5 in *Algorithm 1*). The individual round time of mobile device i can be obtained by $T_B + T_{T,i} + T_{U,i}$. At this time, because the selected number of mobile devices have been selected is unknown, the individual round time is calculated based on the assumption that mobile device i exploits the entire bandwidth B . After that, if mobile device i^* has any data classes that have not been included in the aggregated data class of mobile devices in $\mathbf{A}_S$ (i.e., $(\bigcup_{i \in \mathbf{A}_S} D_i) - D_{i^*} \neq \emptyset$), mobile device i^* is added to $\mathbf{A}_S$ (line 7 in *Algorithm 1*). In addition, the index k is assigned to mobile device i^* (line 8 in *Algorithm 1*), and mobile device i^* is removed from $\mathbf{A}_P$. This procedure is repeated until the ratio between the number of aggregated data classes of the selected mobile devices and the number of whole data classes is greater than or equal to a pre-defined lower bound θ_C (i.e., $\frac{\prod_{i \in \mathbf{A}_S} |D_i|}{K} \geq \theta_C$) or until all the mobile devices are examined (i.e., $\mathbf{A}_P = \emptyset$).

To guarantee a sufficient number of trained data per round, more mobile devices are selected. Specifically, to prevent the round time from becoming excessively long, mobile device i^* with the shortest individual round time $T_{L,i}$ is selected from $\mathbf{A}_P$ (line 13 in *Algorithm 1*). Accordingly, $\mathbf{A}_S$ and $\mathbf{A}_P$ are updated by adding and removing mobile device i^* to $\mathbf{A}_S$ and from $\mathbf{A}_P$, respectively (lines 14-15 in *Algorithm 1*). This procedure is repeated until the number of

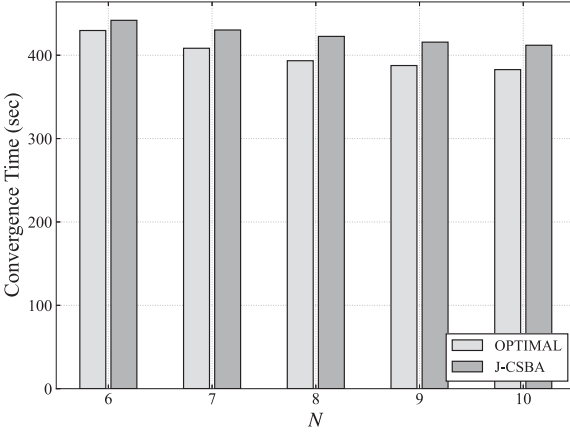


Fig. 4. Optimal versus J-CSBA.

trained data in the selected mobile devices is equal to θ_M or $\mathbf{A}_P = \emptyset$.

After selecting mobile devices, appropriate bandwidths should be allocated to them based on their individual round time T_r . Note that, because the latency of a round T is determined by the slowest mobile device selected in that round, more bandwidth should be allocated to such mobile devices with larger indexes. Therefore, $\frac{T_{I,k}}{\sum_k T_{I,k}} B$ is allocated to the mobile device with index k .

Algorithm 1 has three loops and thus the computational complexity of Algorithm 1 is decided by the highest complexity among these loops. The maximum number of iterations for the first and the second loops is determined by the number of mobile devices (i.e., N). Meanwhile, the maximum number of iterations for the last loop is affected by the number of selected mobile devices. Therefore, the complexity of Algorithm 1 is decided by the first or second loop, which is given by $O(N)$ (i.e., linear complexity). That is, Algorithm 1 can be easily implemented in real systems without any high complexity overhead.⁷

7 EVALUATION RESULTS

For performance evaluation, we compare J-CSBA, with the following four schemes: 1) ALL where all mobile devices are selected and bandwidth is evenly allocated to them; 2) RAND where M mobile devices are randomly selected and bandwidth is randomly allocated to them; 3) FASTEST where one mobile device with the shortest parameter return time is selected and whole bandwidth is allocated to it; and 4) M -MOBI [23] where M mobile devices with the shortest parameter return time are selected and bandwidth is evenly allocated to them; and 5) DIST [28] where several mobile devices with the shortest parameter return time are selected to make their aggregated data distribution near IID and bandwidth is randomly allocated to them. Meanwhile, the convergence time to achieve a certain accuracy of the model (i.e., 80%) is used as one of performance measures of J-CSBA.⁸ In addition, the average number of trained data in

TABLE 1
Running Time (Unit: msec) of Optimal and J-CSBA

N	10	30	50	70	90
OPTIMAL	84.3649	4.9×10^6	N/A	N/A	N/A
J-CSBA	0.0803	0.1052	0.1308	0.1557	0.1807

the round and the ratio between the number of data classes learned in the round and the number of whole data classes, ψ_M and ψ_C , should be maintained above a certain number, these are used as performance measures of J-CSBA as well.

The default parameter settings are as follows. The total number of mobile devices N is set to 30 [29] and M is set to 9. Each mobile device has 1000 images in 2 classes [30] of CIFAR-10. The lower bounds on the ratio between the number of data classes and on the number of trained data, θ_C and θ_M , are set to 0.7 and 3000, respectively. The model size W is 2.3 MBytes. The model is composed of six 3×3 convolutional layers, a fully connected layer with 512 units and ReLu, and a final Softmax output layer. Each convolutional layer has 32, 32, 64, 64, 128, and 128 filters and uses ReLu. In addition, every two convolutional layers are followed by 2×2 maxpooling. We assume that the computing time $T_{T,i}$ follows a normal distribution with a mean of 11 seconds⁹ and standard deviation of 1.5 seconds [23].¹⁰ We consider one of the default frame structures and bandwidth configurations in the 3GPP standard (i.e., frame structure type 2 [31] with 10 MHz channel bandwidth [32]). In addition, the noise power N_O and transmission power density of mobile device P_M are set to -102 dBm/MHz and 7 dBm/MHz, respectively [33]. It is assumed that the channel gain H follows a binomial distribution, which is a discrete version of a normal distribution, where the mean and standard deviation are -98.2 dB/MHz and 2 dB/MHz, respectively [34].

7.1 Optimal Versus J-CSBA

In this analysis, the optimal solution has been obtained by means of the brute-force search. Fig. 4 shows the effect of the number of mobile devices, N , on the convergence time. It can be found that J-CSBA can achieve a convergence time comparable to that of an optimal solution. This can be explained as follow. First, J-CSBA selects the mobile devices to maintain the ratio between the number of data classes learned in the round and the number of whole data classes above a certain level θ_C . Also, J-CSBA maintains a sufficient number θ_M of trained data in each round. Moreover, more bandwidth is allocated to mobile devices with lower computing power and worse channel condition to avoid the straggler problem, which contributes to the reduction of the convergence time.

7.2 Running Time of J-CSBA

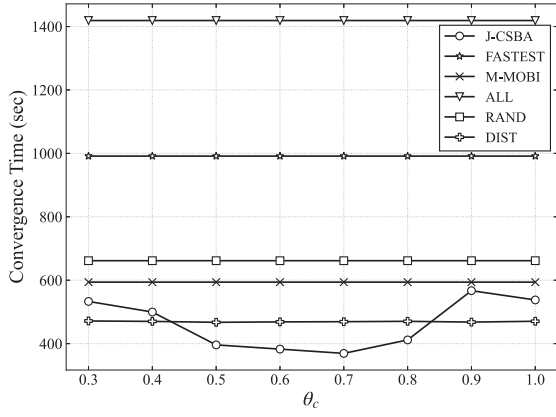
Table 1 shows the running time for the optimal solution and J-CSBA according to the number of mobile devices N . We have run the brute-force search for the optimal solution and

7. Since the number of iterations of Algorithm 1 is constrained by a specific number, its termination condition is always satisfied.

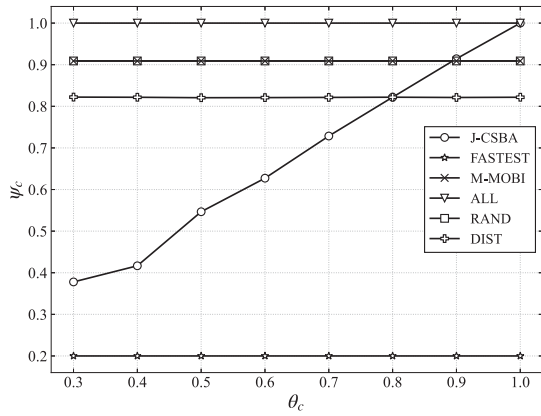
8. The convergence time is measured based on the approximated number of rounds for convergence, and this number is derived by the regression results of extensive simulations [33].

9. We measured the average training latency by using Intel i7-8700 CPU laptop without GPU.

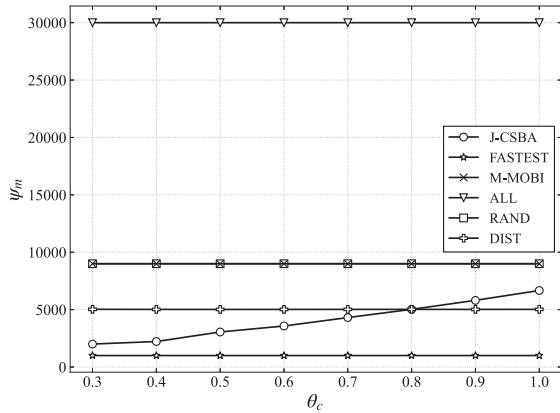
10. With this assumption, $P[C'_i|C_i]$ can be derived because the computing time is proportional to the computing power.



(a) Average latency to achieve a certain accuracy.



(b) Average ratio between the number of data classes learned in the round and the number of whole data classes.



(c) Average number of trained data.

Fig. 5. Effect of the lower bound on the ratio between the number of data classes learned in the round and the number of whole data classes.

J-CSBA by using Intel i9-9880H CPU and 16G RAM. As shown in Table 1, it can be found that the running time of J-CSBA increases as N increases. This is because the number of iterations of J-CSBA increases proportionally to N . However, even when N is large (e.g., 90), the running time is not quite long (e.g., 0.1807 msec), which implies that J-CSBA can be implemented in real systems in a scalable manner.

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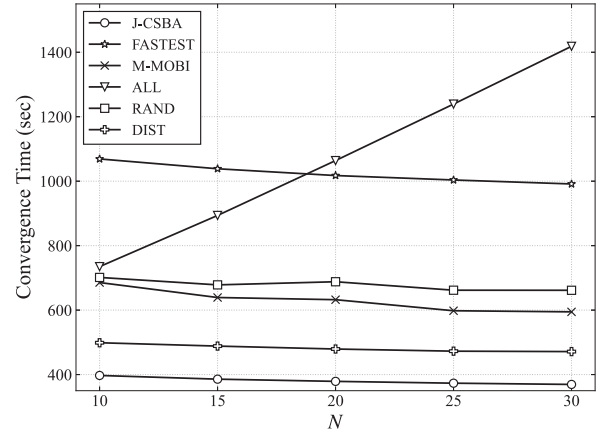


Fig. 6. Effect of the total number of mobile devices on the average latency to achieve a certain accuracy.

On the other hand, the number of iterations for the brute-force search exponentially increases according to N . Thus, only when N is 30, the running time is 4.9×10^6 msec. In addition, due to a memory shortage issue, we could not run the brute-force search for the optimal solution when N is larger than 30.

7.3 Effect of θ_C

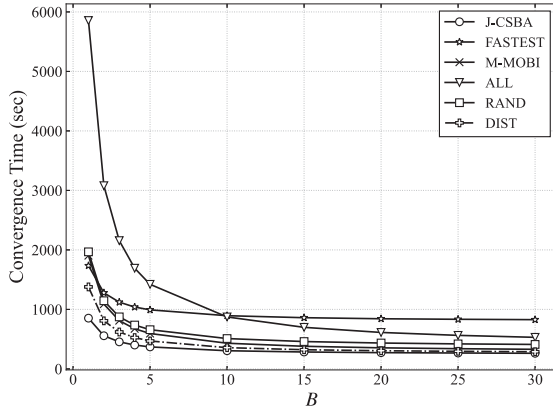
Fig. 5 shows the effect of the lower bound θ_C on the ratio between the number of data classes learned in the round and the number of whole data classes. From Fig. 5a, it can be found that the convergence time of J-CSBA decreases with the increase of θ_C from 0.3 to 0.7. This can be explained as follows. First, more data classes are used to update the parameters in each round (see Fig. 5b). In addition, to maintain the average ratio ψ_C between the number of data classes learned in the round and the number of whole data classes above a desired level (see Fig. 5b), J-CSBA selects more mobile devices in each round, which allows more trained data to be collected (see Fig. 5c).

From Fig. 5a, it can also be seen that the convergence time of J-CSBA tends to increase as θ_C increases from 0.7 to 1. This can be explained as follows. When too many mobile devices are selected, it is likely that undesirable mobile devices with low computing power and/or poor channel conditions are included. As a result, an increased convergence time is expected. Note that because the other schemes do not consider θ_C to select the mobile devices, their ζ_L , ψ_M , and ψ_C do not change regardless of θ_C as shown in Fig. 5.

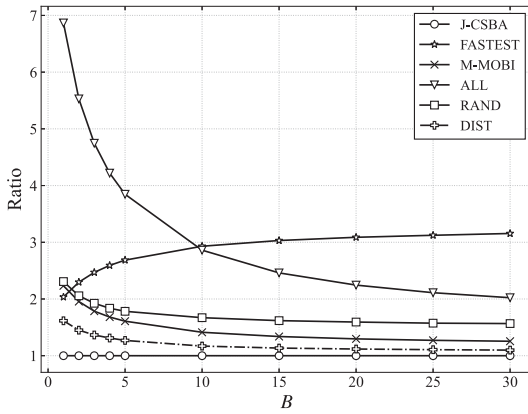
Because DIST selects several mobile devices by considering their parameter return time and data distributions as in J-CSBA, its convergence time can be lower than that of J-CSBA with non-optimal settings (e.g., too low or high value of θ_C) as shown in Fig. 5a. However, since DIST allocates randomly the bandwidth to mobile devices, its convergence time cannot be comparable to that of J-CSBA with the optimal setting of θ_C (i.e., 0.7).

7.4 Effect of N

Fig. 6 shows the effect of the total number N of mobile devices on the convergence time when θ_C is set to 1. From Fig. 6, it can be seen that the convergence time of all the schemes except ALL and RAND decreases with increasing N . This



(a) Average latency to achieve a certain accuracy.



(b) Relative latency ratio of comparison schemes based on that of J-CSBA.

Fig. 7. Effect of the total bandwidth.

can be explained as follows. With the increase in the value of N , more mobile devices with better performance can be expected in device selection. In other words, because J-CSBA, M-MOBI, and FASTEST select mobile devices with better performance (e.g., shorter parameter return time, more data classes, etc.), their average latency to achieve a certain accuracy decreases. Meanwhile, because RAND selects randomly mobile devices and then randomly allocates bandwidths to them, its convergence time is rarely affected by N . In addition, because the latency of a round is determined by the slowest mobile device and all the mobile devices are selected in ALL, the convergence time significantly increases. This result indicates that mobile devices with poor performance (i.e., stragglers) should be excluded in an appropriate manner.

7.5 Effect of B

Figs. 7a and 7b show the effect of the total bandwidth B on the convergence time and the relative ratio of comparison schemes to J-CSBA, respectively. As shown in Fig. 7a, J-CSBA can achieve the lowest convergence time regardless of B . This is because J-CSBA allocates bandwidth such that it is inversely proportional to the expected individual round time of the selected mobile devices (see line 20 in *Algorithm 1*). In addition, from Fig. 7a, it can be found that the

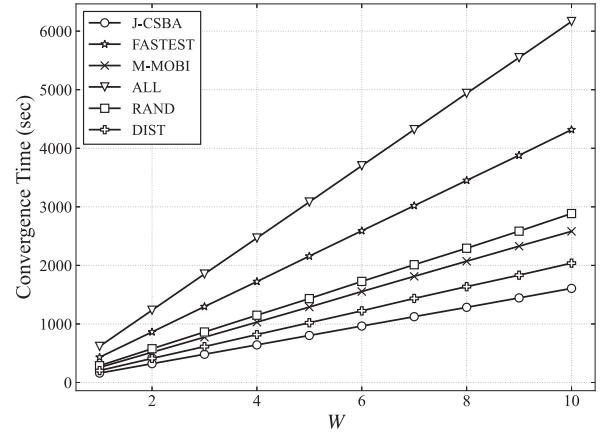


Fig. 8. Effect of the parameter size of the model on the average latency to achieve a certain accuracy.

convergence time of all the schemes decreases with increasing B . This is because a large B means that the selected mobile devices can receive the parameter and transmit the updated one within a short duration.

Meanwhile, from Fig. 7b, it can be found that J-CSBA can achieve higher performance gain compared to the other schemes except FASTEST when limited bandwidth B is considered. This can be explained as follows. When B is large, each mobile device may have sufficient bandwidth regardless of the bandwidth allocation method. In such a situation, the effectiveness of the bandwidth allocation method in J-CSBA can relatively decrease. On the contrary, the effectiveness of J-CSBA becomes apparent when a limited bandwidth is considered.

7.6 Effect of W

Fig. 8 shows the effect of the parameter size W of the model on the convergence time. Intuitively, it can be found that the convergence time of all the schemes increases with increasing W . Interestingly, it can be found that the increasing rate of the convergence time of J-CSBA is the smallest among the comparison schemes. This is because J-CSBA adaptively allocates more bandwidth to mobile devices with lower computing power and worse channel conditions. In so doing, the latency of a single round can be minimized. This result indicates that J-CSBA can achieve more performance gain with a larger size of the model.

8 CONCLUSION

In this paper, we proposed J-CSBA that jointly selects appropriate mobile devices and allocates relevant amounts of bandwidth to them by considering their data information, computing power, and channel gain. We derived the optimal mobile devices and bandwidth allocation by formulating CMDP problem. In other words, based on CMDP, J-CSBA can operate with the optimal policy that can minimize the average time of rounds while maintaining the number of trained data and number of classes of the selected mobile devices in a single round above certain numbers. Evaluation results demonstrated that J-CSBA with the optimal policy can reduce the convergence time by up to 49% in comparison with a random scheme. In addition, it was found that J-CSBA can achieve more

performance gain compared to other schemes when the given bandwidth is small and the model size is large. In our future work, we will investigate an incentive mechanism to encourage mobile devices to consume more computing power for FL.

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